AI & ML Internship – Task 3

Model Validation, Overfitting Control & Hyperparameter Tuning

Objective

The objective of this task is to understand model validation techniques, detect overfitting and underfitting, apply cross-validation, perform hyperparameter tuning, and compare machine learning models using regression metrics.

Technologies Used

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, scikit-learn

Dataset Used

The California Housing dataset from scikit-learn was used for house price prediction.

Concepts Covered

Train-Test Split, Feature Scaling, Linear Regression, Ridge Regression, Decision Tree Regression, Overfitting Detection, Cross Validation, GridSearchCV, Hyperparameter Tuning, RMSE Evaluation, R^2 Score Evaluation, and Model Comparison.

Project Workflow

1. Import Required Libraries
2. Load Dataset
3. Data Preprocessing
4. Train-Test Split
5. Model Training
6. Cross Validation
7. Hyperparameter Tuning
8. Model Evaluation
9. Model Comparison

Best Model

The Tuned Decision Tree Regressor achieved the best performance with the lowest RMSE and highest R^2 score. Hyperparameter tuning improved generalization performance and reduced overfitting.

Key Learning Outcomes

- Understanding overfitting and underfitting
- Applying cross-validation correctly
- Performing hyperparameter tuning professionally
- Evaluating regression models using RMSE and R^2
- Building generalizable machine learning models
- Following industry-standard ML workflow

Conclusion

This task successfully demonstrated model validation, overfitting control, and hyperparameter tuning using machine learning regression models. Among all tested models, the Tuned Decision Tree Regressor provided the best prediction performance.

Final Results

Model	RMSE	R^2 Score
Linear Regression	0.745581	0.575788
Ridge Regression	0.745554	0.575819
Tuned Decision Tree	0.645430	0.682099